

Astrodynamics I: Final Exam

June 1, 2016 11:05-12:45PM (100 minutes)

Closed Books/Notes

Use the following formula without proof if necessary:

$$\bullet \quad \vec{e} = \frac{1}{\mu} \left[\left(v^2 - \frac{\mu}{r} \right) \vec{r} - (\vec{r} \cdot \vec{v}) \vec{v} \right], \quad e = \sqrt{1 + \frac{2Eh^2}{\mu_E^2}}$$

1. (20) Given $\vec{r} = -\frac{1}{\sqrt{2}}\hat{I} + \frac{1}{\sqrt{2}}\hat{J}$ and $v = \frac{1}{2}\hat{J}$, determine the orbital elements and sketch the orbit. Use $\mu = 1$.

2. (20) Given the following set of orbital elements,

$$(a = 1.2175, e = 0.5195, i = 45^\circ, \Omega = 270^\circ, \omega = 104.82^\circ, \nu = 200.4418^\circ)$$

determine its position and velocity vector in the geocentric equatorial coordinate system. Use $\mu = 1$.

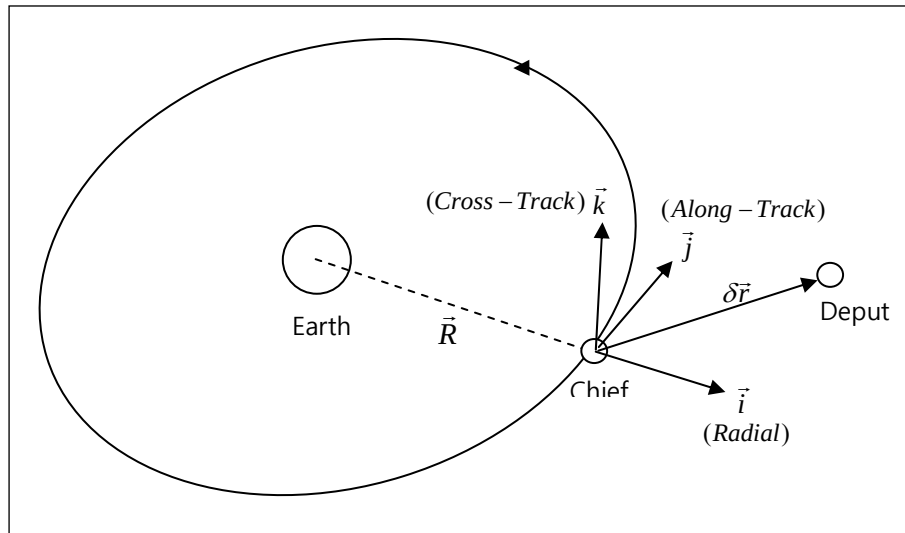
3. (20) Consider a local-vertical-local-horizontal xyz-frame attached to a circular reference orbit whose constant orbital frequency is given by $\vec{\omega} = \omega\vec{k}$. Let the position of deputy spacecraft with respect to the circular reference orbit (or the chief spacecraft in the figure) be given by

$$\vec{r} = \vec{R} + \delta\vec{r} = (R+x)\vec{i} + y\vec{j} + z\vec{k}$$

- A. (15) Determine the velocity and acceleration vector of the deputy spacecraft in terms of $(\vec{i}, \vec{j}, \vec{k})$ -component.
- B. (5) Use the two-body equations of motion

$$\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r}$$

to derive the equations of motion in terms of $(\vec{i}, \vec{j}, \vec{k})$ -component. Here μ is the gravitational parameter of the Earth and $r = \sqrt{(R+x)^2 + y^2 + z^2}$ is the distance of the deputy spacecraft from the Earth center.



4. (20) Discuss the following succinctly and precisely.
 - A. (10) Apparent Solar Time, Mean Solar Time, Universal Time, Equation of Time
 - B. (10) Mean Sidereal Time, Apparent Sidereal Time, Equation of Equinoxes

5. (20) Given $\vec{r}_{IJK}, \vec{v}_{IJK}, \vec{r}_{SiteIJK}, \Phi_{gd}, \theta_{LST}$, describe the procedure to compute range (ρ), azimuth (β), elevation (el), and their time rates.